

Problem 10.28

You are given that

$$i^{(12)} = 0.09.$$

Calculate δ .

Solution

The nominal annual rate $i^{(12)} = 0.09$ is convertible monthly, so the monthly effective interest rate is

$$\frac{i^{(12)}}{12} = \frac{0.09}{12} = 0.0075.$$

Therefore, the annual accumulation factor is

$$\left(1 + \frac{0.09}{12}\right)^{12}.$$

The equivalent constant force of interest δ satisfies

$$e^{\delta} = \left(1 + \frac{0.09}{12}\right)^{12}.$$

Taking natural logarithms gives

$$\begin{aligned}\delta &= \ln \left[\left(1 + \frac{0.09}{12}\right)^{12} \right] \\ &= 12 \ln \left(1 + \frac{0.09}{12}\right) \\ &\approx 0.089664178064.\end{aligned}$$

Thus, the equivalent force of interest is

$$\boxed{\delta \approx 8.9664\%}.$$